Experiment 6

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Section/Group: 23BCS_KRG-2_A
Semester: 6th
Date of Submission: 02/03/2026
Subject Name: System Design
Subject Code: 23CSH-314

1- Aim - To design a scalable and highly available YouTube-like distributed video streaming system for large-scale users.

2- Requirements: Functional & Non-Functional A- Functional Requirement

- User Registration & Login
- Video Upload by Creator
- Search Videos
- Watch Videos in multiple resolutions (360p, 720p, 1080p, 4K)
- Like a Video
- Comment on a Video
- View Like count & Comments

B- Non-Functional Requirement

- Scalability: 500M+ Users
- High Availability >>> Consistency
- Low Latency (<100ms interactions)
- Zero / Negligible Buffering
- High write throughput for Likes & Comments

3- Core-entities of System

- Users
- User Metadata
- Videos
- Video Metadata
- Manifest File
- Video Segments
- Likes
- Comments

4- API endpoint creation

1. **Video Upload**
   1. POST Call: /video/upload
   2. POST Call: /video/{video_id}/metadata

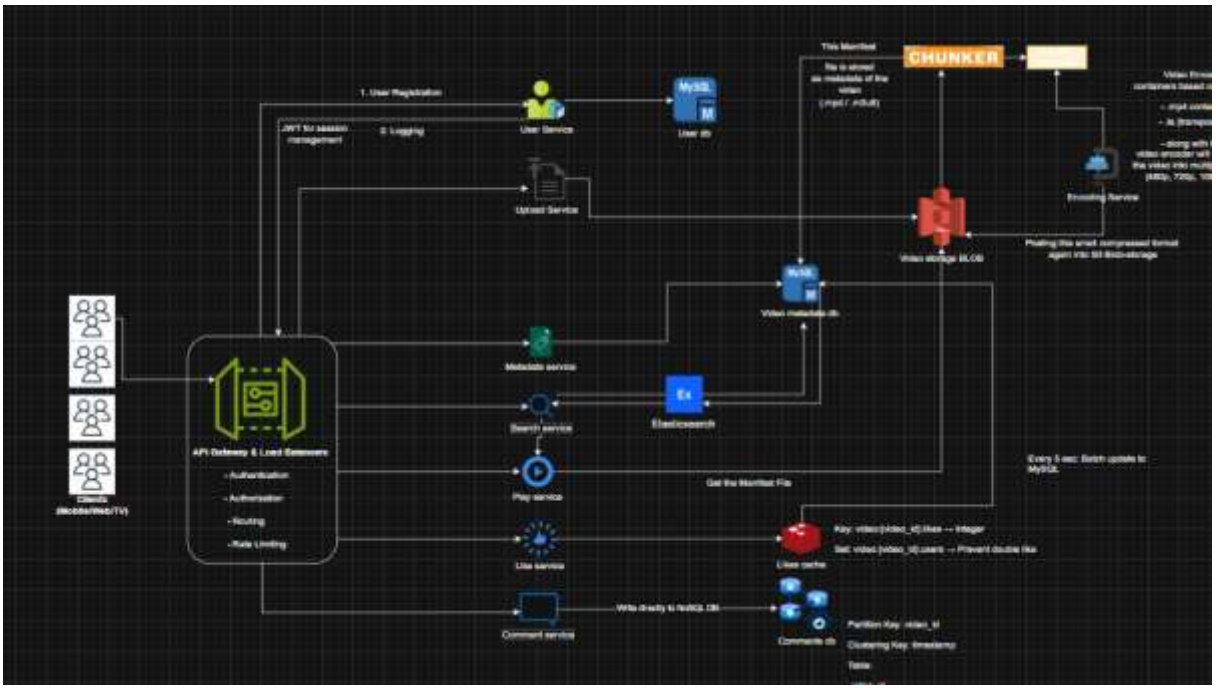
2. **Streaming**

   1. GET Call: /video/{video_id}
   2. GET Call: /video/{video_id}/manifest.mpd
   3. GET Call: /video/{video_id}/manifest.m3u8
   4. GET Call: /video/{video_id}/segment/{segment_id}

3. **Likes**

   1. POST Call: /video/{video_id}/like
   2. DELETE Call: /video/{video_id}/like
   3. GET Call: /video/{video_id}/likes/count

4. **Comments**

   1. POST Call: /video/{video_id}/comment
   2. GET Call: /video/{video_id}/comments?cursor=xyz

5- Low-Level Design